SHUHAN (Alice) AI

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EDUCATION

PhD, School of Education, University of California, Los Angeles, CA 2023.09 – 2027.06 (expected)
- Major: Higher Education & Organizational Change (HEOC)
- Advisor: Kevin Eagan

M.S., Department of Statistics, University of California, Los Angeles, CA 2025.09 – 2027.06 (expected)
- Major: Statistics and Data Science
- Advisor: Qing Zhou

M.S.Ed, University of Pennsylvania, Philadelphia, PA 2021.09 – 2022.12
- Major: Higher Education, Graduate School of Education

BA, Northeast Normal University, Jilin, China 2017.09 – 2021.06
- Major: Education, School of Education Science

ACADEMIC APPOINTMENTS

Teaching Assistant
- EDUC230A Introduction to Research Design and Statistics (graduate course) 2025.09 – 2026.03
- EDUC230B: Linear Statistical Models: Analysis of Designed Experiments (graduate course) 2026.04 – 2026.06

Graduate Student Researcher, UCLA, CA
School of Education and Information Studies 2023.09 – 2025.06
- Principal Investigator: Dr. Kevin Eagan
- Current Project: The Effects of BUILD Participation on the Development of Science Identity and STEM Aspirations

Journal Editor, UCLA, CA
InterActions: UCLA Journal of Education and Information Studies 2024.01 – 2025.06
- Coordinate, manage journal operations, and communicate with the Journal Advisory Board and Research Groups.
- Supervise manuscript submission, peer-review processes, and editorial workflows to support authors and reviewers.
- Maintain and update the journal's eScholarship platform, ensuring accuracy for all published content.

RESEARCH INTERESTS

STEM; Higher Education; Multilevel Modeling; Causal Inference; Network Analysis; Survival Analysis

AWARDS & FELLOWSHIPS & FUNDINGS

- UCLA William and Evelyn Scholarship, $30,000/year 2023, 2025
- UCLA Division of Graduate Education Fellowship, 15,000/year 2024, 2025
- UCLA Graduate Summer Research Mentorship Award (GSRM), $6,000/year 2024, 2025
- ASHE Graduate Student Travel Scholarship $400 2025
- Excellent Student, The First Prize Scholarship, CNY 4,000/year 2018, 2019, 2020
- Outstanding Graduates Award, Social Practice Award, CNY 3,000 2021
- Fundamental Research Funds for the Central Universities, CNY 5,000 2018
- Scholarship for Exchange Student, CNY 20,000 2018

PUBLICATIONS

Ai, S., Eagan, K., Wu, J. (under review). Fueling Ambition: Graduate Degree Aspirations Among Women of Color in STEM. *CBE – Life Science Education.*

Ai, S. & Eagan, K., (under review). From Club to College: The Impact of Science Clubs/Groups on High School Students' STEM Major Choice. *Research in Higher Education.*

Ai, S., Eagan, K., Ross-Olivares, J. P., & Jiang, T. (under review). Can community cultural wealth maintain undergraduate women of color toward STEM careers? *Journal of Diversity in Higher Education.*

Jiang, T., **Ai, S.,** Jiao, C., Li, K. (under review). Revealing Hidden Contributions on Research Data: The Potential of Data Paper. *Scientometrics.*

Ai, S. (2024). Empowering Online Teaching: A System Review of Online Instructors' Professional Development in Higher Education. *InterActions: UCLA Journal of Education and Information Studies, 19(1).*

Omery, F., Montes-Diaz, I., **Ai, S.** (2023). City Fiscal Conditions. National League of Cities. https://www.nlc.org/resource/city-fiscal-conditions-2023/

Omeyr, F. & **Ai, S.** (2023). State of the Cities. National League of Cities. https://www.nlc.org/resource/state-of-the-cities-2023/

CONFERENCE PAPER PRESENTATION

Ai, S., Eagan, K., & Wu, J. (2025). Fueling Ambition: Graduate Degree Aspirations Among Women of Color in STEM. Paper presentation at the annual meeting of the Association for the Study of Higher Education, Denver, CO.

Ai, S., Eagan, K. (2025). What Factors Promote High School Students' Development of Science Identity and STEM Major Aspiration? An Analysis of HSLS:09 Data. Roundtable presentation at the annual meeting of the American Education Research Association, Division J: Postsecondary Education, Denver, CO.

Ai, S. (2025). A Critical Race Analysis of Black and Hispanic Female Graduate Students Enrollment Trends from 2012 to 2021. Roundtable presentation at the annual meeting of the American Education Research Association, Division J: Postsecondary Education, Denver, CO.

Ai, S., Rios, M., Olivares, J., & Jiang, T. (2024). Can Community Cultural Wealth Guide Undergraduate Women of Color Toward STEM Careers? Examining HERI Data. Paper presentation at the annual meeting of the Association for the Study of Higher Education, Minneapolis, MN.

Ramos, H., Eagan, K., & **Ai, S**. (2024). BUILDing Science Identity: Exploring the Efficacy and Moderation Effects of a Nationally Funded Intervention. Paper presentation at the annual meeting of the Association for the Study of Higher Education, Minneapolis, MN.

Jiang, T., **Ai, S.,** & Li, K. (2024). Revealing Contributions on Research Data: The Potential of Data Paper. Paper presentation at the MET/STI 2024 Workshop of the Association for Information Science and Technology (Virtual).

Ai, S. (2024). STEM Curriculum as a Catalyst: Narrowing Racial and Structural Inequalities. Roundtable presentation at the annual meeting of the American Education Research Association, Division L: Educational Policies and Politics, Philadelphia, PA.

Liu, Xi., & **Ai, S.** (2024). Improving Affect Detections for K-12 Online Math Learning Using Time-related Variables. Paper presentation at the annual meeting of the American Education Research Association, Division C: Learning and Instruction, Philadelphia, PA.

PARTICIPATION IN FUNDED GRANTS

National Institute of General Medical Sciences (Grant No. U54GM119024)
Graduate Student Researcher 2023.09-2025.06
Diversity Program Consortium Coordination and Evaluation Center at UCLA 2015-2025
Funded by National Institutes of Health (NIH)

UCLA Graduate Summer Research Mentorship award (GSRM 2025)
Principal Investigator 2025.06-2025.09
Race, Identity, and Context: A Multilevel Modeling Analysis of Senior Women's STEM Graduate Degree Aspirations
Funded by UCLA Fellowship: $6,000

UCLA Graduate Summer Research Mentorship award (GSRM 2024)
Principal Investigator 2024.06-2024.09
Factors Promote High School Students' Science Identity and STEM Major Aspiration? Evidence from HSLS:09
Funded by UCLA Fellowship: $6,000

PEER REVIEWER

Journal of Women and Minorities in Science and Engineering
Annual Meeting of the Association for the Study of Higher Education
Annual Meeting of the American Education Research Association

RELEVANT RESEARCH EXPERIENCE

BUILDing Science Identity: Exploring the Efficacy and Moderation Effects of a Nationally Funded Intervention
GSR Researcher Assistant 2023.09 – present

- Funded by National Institutes of Health (NIH). Conducted quasi-experimental research to assess the efficacy of NIH's BUILD initiative in promoting science identity among underrepresented undergraduate students in STEM.
- Applied advanced statistical methods, including logistic regression, propensity score modeling, and inverse probability weighting, to evaluate program effectiveness.
- Delivered actionable insights to inform policies on mentoring, research opportunities, and financial support to enhance student persistence and STEM degree attainment.

Can Community Cultural Wealth Guide Undergraduate Women of Color Toward STEM Careers?
Principal Investigator 2024.02 – present

- Conducted quantitative analysis using longitudinal data from HERI's Freshman and College Senior Surveys to explore the impact of Community Cultural Wealth (CCW) on STEM career aspirations among women of color.
- Operationalized and analyzed CCW dimensions (e.g., aspirational, familial, social, navigational, resistant) to evaluate their influence on senior undergraduate students' STEM career goals.
- Applied logistic regression to assess the predictive power of CCW dimensions on STEM career aspirations.

Shaping Science Identity: Parental Influence and STEM Aspiration Among High School Students
Principal Investigator 2024.07 – present

- Conducted quantitative analysis with High School Longitudinal Study (HSLS:09) data involving 23,503 samples.
- Explored the role of parental involvement (e.g., museum visits, science discussions) on students' science identity.
- Applied nested regression models and logistic regression to assess the impact of science identity on STEM aspirations.

Feature Engineering and Affect Detection for K-12 Online Math Learning
Co-Principal Investigat; Code in Github 2022.09 – 2022.12

- Conducted feature importance analyses with XGBoost, Decision Tree, and Logistic Regression.
- Used 10-fold student-level batch cross-validation to prevent model overfitting.
- Built data mining models, compared different Kappa value, AUC, and Accuracy.
- Analyzed Correlation Matrix by using heatmap in python.
- Paper presented in 2024 AERA conference.

Social Network Analysis on Co-authorship in Community College Enrollment
Researcher; Code in Github 2022.09 – 2022.12

- Conduct the two-mode to one-mode transformation retaining co-authors.
- Used Key Actor Analysis and plotted the weighted sociogram with the top 5% of influential actor.
- Created two-mode interactive network visualization with networkD3.

WORK EXPERIENCES

National League of Cities (Full-time), Washington, D.C.
Research and Data Consultant (40hr/week), Center for Research and Data Analysis 2023.01 – 2023.09

- Integrated natural language processing into the coding of the APRA categories in STATA and Python.
- Wrote NLC's State of the Cities (SOTC) reports and performed data analysis.
- Supported the cleaning of the Preemption Project data in STATA.
- Collected data and conducted report for NLC's City Fiscal Conditions.

Wharton Online, Philadelphia, PA
Teaching Assistant (25hr/week), Wharton Executive Education 2021.09 – 2023.06

- Written weekly online global learning reports for major clients.
- Collaborated with the Coursera, Emeritus, Canvas team to refine online course structures.
- Monitored 30+ MOOCs, 2.3 million learners, and 15+ Business to Business (B2B) specializations.
- Engaged in course building for new and existing B2B clientele, evaluating course design and student interaction.

COLLEGE ACTIVITIES

Summer Institutes in Computational Social Science (SICSS-UCLA), Los Angeles, CA
Participant 2024.06 – 2024.07

- Gained comprehensive training in advanced computational methods and their applications in social science research.
- Collaborated on interdisciplinary projects, employing techniques such as network analysis, and machine learning.
- Enhanced proficiency in utilizing statistical software and programming languages for data analysis, contributing to innovative solutions in higher education research.
- Applied the Potential Outcome Model for causal inference, developing skills in estimating treatment effects and addressing confounding factors.

The Walnut Street Club, University of Pennsylvania, GSE, Philadelphia, PA
Co-founder 2022.05 – 2023.06

- Ideated, planned, and co-wrote the curriculum for eight-week online training boot camp, taught to over 200 students.
- Cooperated with innovative schools, Penn Catalyst, Montessori Charter School share successful education practice.
- Hosted thematic education discussion, such as Online Education, STEM Education, SEL, and educational policy.
- Compiled and shared paper digest from Higher Education Policy, American Journal of Education, etc.

Penn China Education Summit (PCES), Philadelphia, PA
Education Technology Panel Director (20h/week) 2021.12 – 2022.05

- Built connections among 20+ renowned scholars and 20+ innovative practitioners from China, U.S., U.K., etc.
- Hold 2 forums and 2 roundtable discussions, attracted 860 audience registrations, 1380 viewers in total.
- Produced education podcasts and recorded videos for media outreach.

PERSONAL SKILLS

Statistical Analyses	R; R Markdown; Mplus; Python; STATA; SPSS; FlexMIRT; RapidMiner; Qualtrics
Statistical Methods	Regression; Causal Inference; Social Network Analysis, Data Mining
Languages	English (professional proficiency), Mandarin (native)
Sports	National Collegiate Table Tennis Association Member, UCLA School Team (NCTTA)
Arts	Chinese Traditional Calligraphy; Oil Painting

REFERENCES

Professor, Qing Zhou
Department of Statistics & Data Science
Email: zhou@stat.ucla.edu

Associate Professor, Kevin Eagan
School of Education and Information Studies
Email: keagan@ucal.edu

Professor, Manuel S. Gonzalez Canche
Graduate School of Education, University of Pennsylvania
Email: msgc@upenn.edu